

Problem 4

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Problem 1 Find the derivatives of the following functions:

(a) $f(x) = x^{e^x} + 7x$

Explanation. $f'(x) = \frac{d}{dx} (x^{e^x}) + \frac{d}{dx} (7x) = \frac{d}{dx} (x^{e^x}) + 7$. So the real problem is to find $\frac{d}{dx} (x^{e^x})$.

We will use logarithmic differentiation.

First, we take natural logarithm of both sides of the equation $g(x) = x^{e^x}$.

$$\ln(g(x)) = e^x \ln(x).$$

Now, we differentiate both sides.

$$\frac{g'(x)}{g(x)} = e^x \ln(x) + \frac{e^x}{x}.$$

Next, we solve for $g'(x)$.

$$g'(x) = g(x) \left(e^x \ln(x) + \frac{e^x}{x} \right).$$

Now, we substitute.

$$g'(x) = x^{e^x} \left(e^x \ln(x) + \frac{e^x}{x} \right).$$

ALTERNATIVE APPROACH

$$\begin{aligned} \frac{d}{dx} (x^{e^x}) &= \frac{d}{dx} (e^{\ln(x)e^x}) \\ &= \frac{d}{dx} (e^{e^x \ln(x)}) \\ &= e^{e^x \ln(x)} \left(e^x \ln(x) + \frac{e^x}{x} \right) \\ &= x^{e^x} \left(e^x \ln(x) + \frac{e^x}{x} \right). \end{aligned}$$

$$\text{Thus, } f'(x) = x^{e^x} \left(e^x \ln(x) + \frac{e^x}{x} \right) + 7.$$

(b) $g(x) = (\ln(x) + 9)^{\sec(x^4)}$

Explanation. We will use logarithmic differentiation.

First, we take natural logarithm .

$$\ln(g(x)) = \sec(x^4) \ln(\ln(x) + 9)$$

Then, we differentiate.

$$\frac{g'(x)}{g(x)} = \sec(x^4) \tan(x^4) 4x^3 \ln(\ln(x) + 9) + \frac{\sec(x^4)}{x(\ln(x) + 9)}$$

We multiply by $g(x)$.

$$g'(x) = g(x) \left(4x^3 \sec(x^4) \tan(x^4) \ln(\ln(x) + 9) + \frac{\sec(x^4)}{x(\ln(x) + 9)} \right)$$

We substitute.

$$g'(x) = (\ln(x) + 9)^{\sec(x^4)} \left(4x^3 \sec(x^4) \tan(x^4) \ln(\ln(x) + 9) + \frac{\sec(x^4)}{x(\ln(x) + 9)} \right)$$

ALTERNATIVE APPROACH

$$\begin{aligned} g'(x) &= \frac{d}{dx} \left((\ln(x) + 9)^{\sec(x^4)} \right) \\ &= \frac{d}{dx} \left(e^{\sec(x^4) \ln(\ln(x) + 9)} \right) \\ &= e^{\sec(x^4) \ln(\ln(x) + 9)} \left(4x^3 \sec(x^4) \tan(x^4) \ln(\ln(x) + 9) + \sec(x^4) \frac{\frac{1}{x}}{\ln(x) + 9} \right) \\ &= (\ln(x) + 9)^{\sec(x^4)} \left(4x^3 \sec(x^4) \tan(x^4) \ln(\ln(x) + 9) + \frac{\sec(x^4)}{x(\ln(x) + 9)} \right). \end{aligned}$$

(c) $f(x) = \frac{(x+1)^5 (\sin(x) + 5)^4}{(x^2 + 5)\sqrt{x-3}}$

Explanation.

$$\begin{aligned}\ln(f(x)) &= \ln(x+1)^5 + \ln(\sin(x)+5)^4 - \ln(x^2+5) - \ln\sqrt{x-3} \\ &= 5\ln(x+1) + 4\ln(\sin(x)+5) - \ln(x^2+5) - \frac{1}{2}\ln(x-3)\end{aligned}$$

Now we differentiate.

$$\frac{f'(x)}{f(x)} = \frac{5}{x+1} + \frac{4\cos(x)}{\sin(x)+5} - \frac{2x}{x^2+5} - \frac{1}{2(x-3)}$$

Then we solve for $f'(x)$.

$$f'(x) = f(x) \left(\frac{5}{x+1} + \frac{4\cos(x)}{\sin(x)+5} - \frac{2x}{x^2+5} - \frac{1}{2(x-3)} \right) \text{ Then we substitute.}$$

$$f'(x) = \frac{(x+1)^5(\sin(x)+5)^4}{(x^2+5)\sqrt{x-3}} \left(\frac{5}{x+1} + \frac{4\cos(x)}{\sin(x)+5} - \frac{2x}{x^2+5} - \frac{1}{2(x-3)} \right)$$

$$(d) \ h(x) = \frac{(x^2-7)^5}{\cos^7(x^2-5)}$$

Explanation. Since $h(x)$ changes sign on its domain, we write :

$$\begin{aligned}\ln|h(x)| &= \ln\left(\frac{|x^2-7|^5}{|\cos^7(x^2-5)|}\right) \\ &= 5 \cdot \ln(|x^2-7|) - 7 \cdot \ln(|\cos(x^2-5)|)\end{aligned}$$

Differentiate both sides with respect to x , (explanation given on the next page):

$$\begin{aligned}\frac{h'(x)}{h(x)} &= 5 \frac{2x}{x^2-7} - 7 \frac{-\sin(x^2-5)2x}{\cos(x^2-5)} \\ h'(x) &= h(x) \left(5 \frac{2x}{x^2-7} - 7 \frac{-\sin(x^2-5)2x}{\cos(x^2-5)} \right) = \\ &= \frac{(x^2-7)^5}{\cos^7(x^2-5)} \left(\frac{10x}{x^2-7} + \frac{14x\sin(x^2-5)}{\cos(x^2-5)} \right) = \\ &= \frac{(x^2-7)^4}{\cos^7(x^2-5)} (10x + 14x(x^2-7)\tan(x^2-5))\end{aligned}$$

EXPLANATION: The function $h(x)$ assumes negative values on some intervals. On those intervals

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$$(\ln(|h(x)|))' = (\ln(-h(x)))' = \frac{-h'(x)}{-h(x)} = \frac{h'(x)}{h(x)},$$

Also, e.g., if $x^2 - 7 < 0$

$$(\ln(|x^2 - 7|))' = (\ln(-(x^2 - 7)))' = \frac{-2x}{-(x^2 - 7)} = \frac{2x}{x^2 - 7} .$$